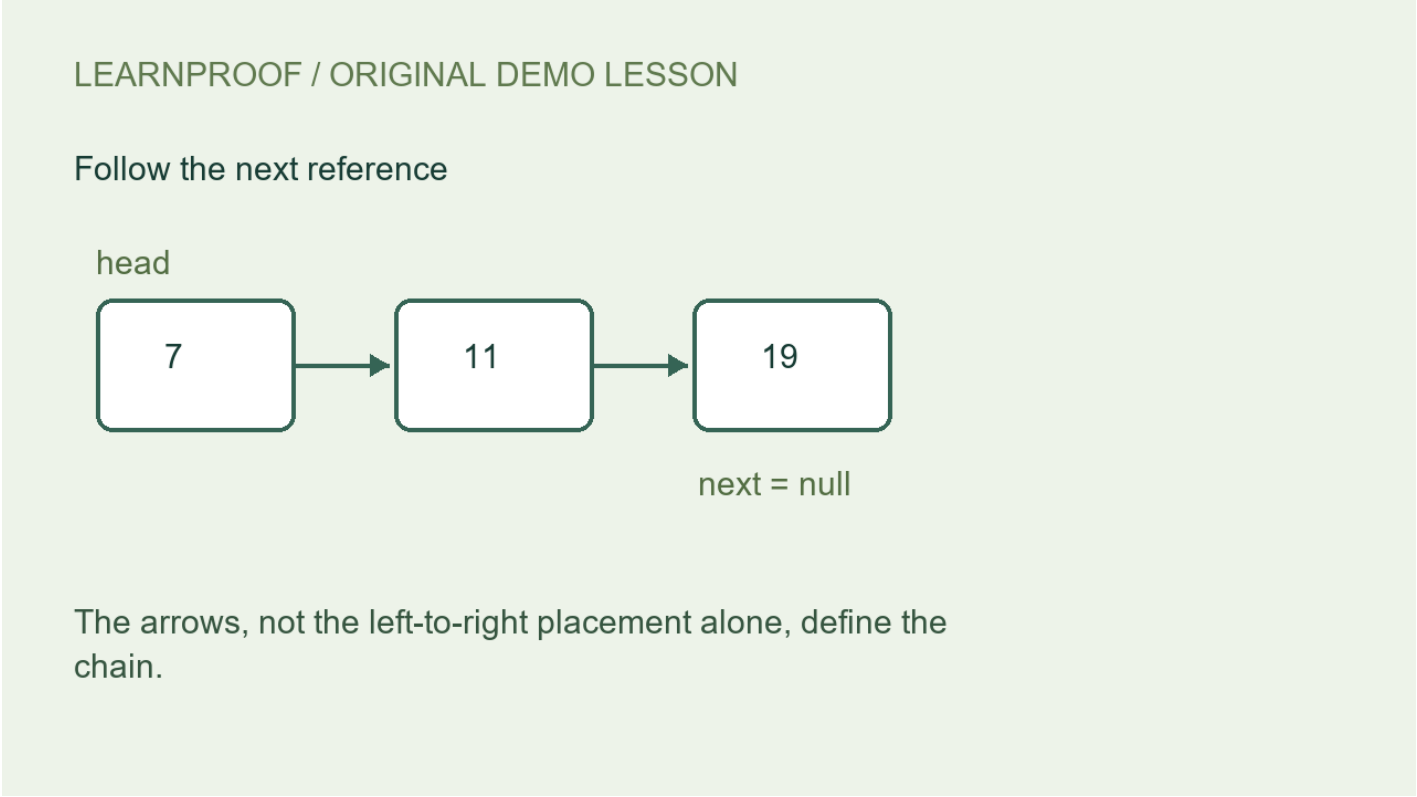
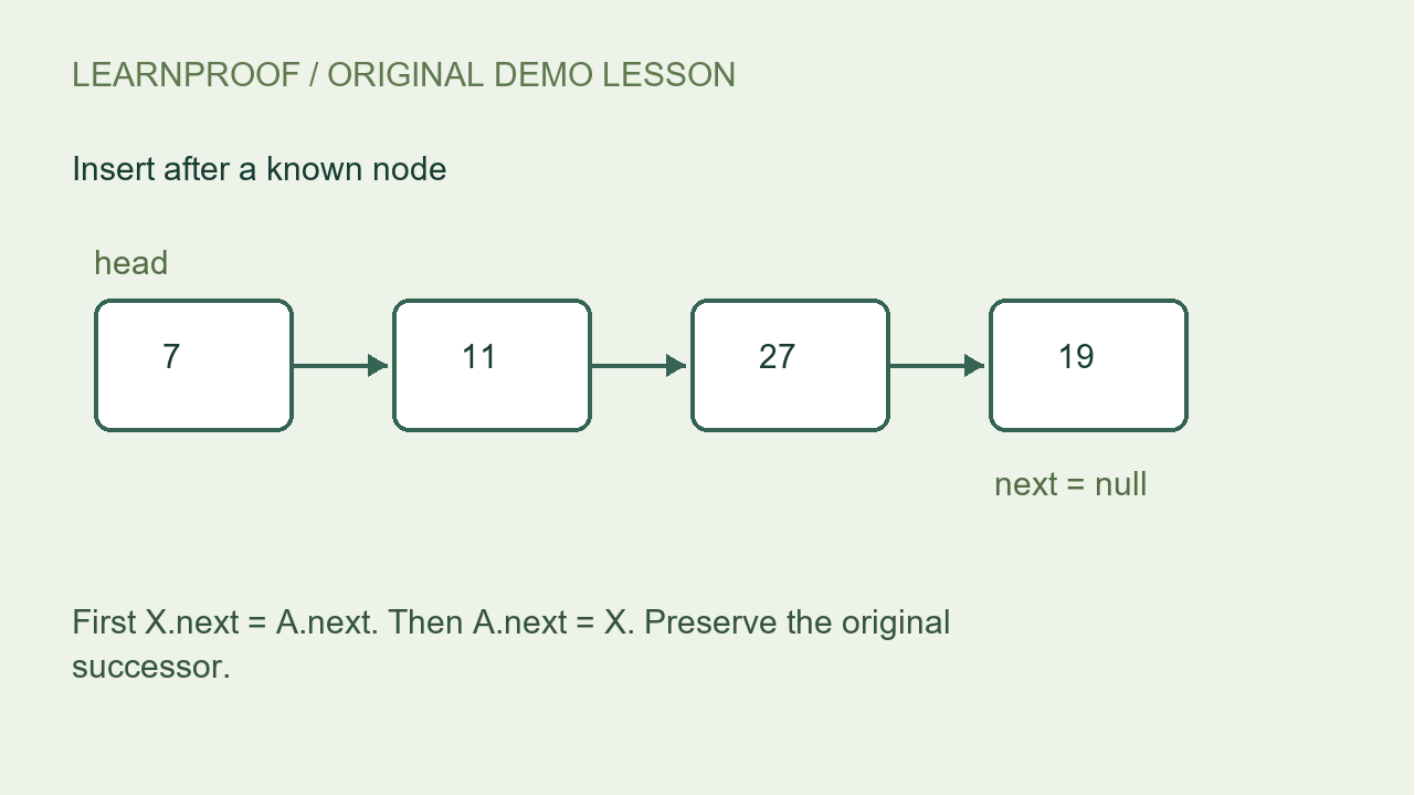
Follow the next reference

Pointers identify nodes. A singly linked list follows next references from head to a null reference. Diagram 1 contains values that must be read from the image; they are intentionally absent from this extracted text.



Insert after a known node

Inserting after an already known node requires two reference updates: X.next = A.next, then A.next = X. Searching for a target value can require n node inspections. Searching is O(n); rewiring is O(1).



Finding is not rewiring

The diagrams show a before and after state. Inspect the direction of each arrow to identify the successor and verify that the original tail remains reachable. A changed picture is evidence; an inferred explanation still requires checking.

